

●PIM GLOBAL LIMITED × QOLLABS LTD

# Pim Protocol *Blueprint*

*Architecture, protocol state machine, consensus and security model, token accounting formulas, core algorithm pseudocode, and parameter tables, extracted and consolidated from the Pim Protocol Technical Whitepaper. Engineer-facing. No marketing language. Every claim below is cited to a Whitepaper section.*

**100,000 TPS**  
THEORETICAL MAX (§4.2)

**16**  
NATIVE PCTS CELL TYPES (§6.7)

**4**  
ADTMS STATES (§6.3)

**5**  
NODE HARDWARE TIERS (APP. B)

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01

## Scope & Method

This Blueprint is a derived technical specification. It reorganizes and condenses claims already made in the Pim Protocol Technical Whitepaper (12 sections, 2 appendices, 42 references) into an engineer-facing reference: architecture, state machine, security model, formulas, pseudocode, and parameter tables, with no explanatory or promotional framing. Every numbered claim below carries a citation to the source

Whitepaper section in the format [WP §x.x] . Where the Whitepaper does not specify a detail this document would otherwise need, that gap is listed explicitly in Section 08 rather than filled with inference.

02

## System Architecture

The protocol composes seven architectural pillars [WP §4.1] into a single stack. Each layer below is independently specified in the Whitepaper and cited at that layer.



### 2.1 Sharding

100 parallel shards, each processing up to 1,000 transactions simultaneously — theoretical maximum 100,000 TPS. DALI (Decentralised Adaptive Learning Intelligence) rebalances shard assignments hourly via Q-learning. Prime Shard Braiding coordinates cross-shard transactions at ~2 ms braiding latency and ~8 s cross-shard finality [WP §5.1] .

### 2.2 State & Storage

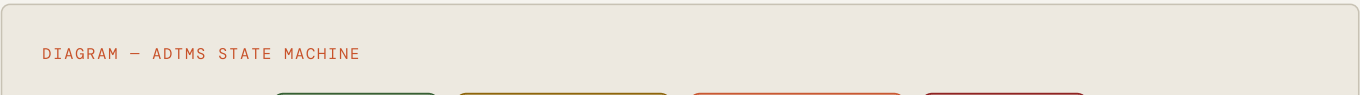
The MMR prunes state to 8.5–12.5 GB/year with compact proofs of 4–8 KB. PYM Cells and QOL UTXOs share a unified MMR structure. MiniCell local storage requires 1–5 GB per node. Cash Mode transactions are stored as "Private Tx" records — output commitments and stealth addresses on-chain; sender, recipient, and amount cryptographically obfuscated [WP §5, storage subsection] .

03

## Protocol State Machine — ADTMS

FOUR-STATE GRADUATED MONETARY CIRCUIT BREAKER

ADTMS v2 governs minting eligibility via a Composite Stress Index (CSI) computed each cycle [WP §6.3] .





Upward transition requires CSI above threshold for 2 consecutive cycles. Recovery (right to left) requires 6 conditions sustained across 4 consecutive cycles, including  $C_{PoEC,t} \geq 0.85$ . No state-skipping permitted on recovery (graduated descent invariant). Transition throttle:  $\leq 2$  state changes per 24h window.

| STATE      | EFFECTIVE SIGNAL      | MINTING                                   |
|------------|-----------------------|-------------------------------------------|
| STABLE     | $1.00 \times EQCF\_t$ | Full rate                                 |
| TIGHTENING | $0.50 \times EQCF\_t$ | Half rate; EQCF drift cap reduced to 1.5% |
| TRANSITION | 0                     | Halted                                    |
| CRITICAL   | 0                     | Halted                                    |

CSI formula and state thresholds are given in Section 06.4. Source: [WP §6.3, App. A] .

04

## Consensus & Security Model

### 4.1 Consensus Mechanism

PoEM (Proof of Entropy Minima): block weight is computed as `BLAKE3-XOF(Block Header, Nonce, Timestamp)`; the minimum-weight candidate wins deterministically, with no probabilistic race [WP §4.1 Pillar II, App. A] . Combined with CSMM (Cross-Shard Merged Mining), DFP (cross-shard finality protocol), and a GHOST-derived fork-choice rule across 100 shards [WP §4.1 Pillar II] .

### 4.2 Sybil Resistance & 51% Attack Surface

Synergic multi-tier node architecture: a 51% attack requires simultaneous compromise of five node tiers across different hardware, software, connectivity, and geography [WP §12.1] . Fork probability at 10,000 simulated nodes: 0.0007% [WP §11] .

### 4.3 Identified Risk Classes and Resolutions

Eight risk classes are identified and formally resolved in the Whitepaper [WP §9] :

#### RISK 9.1

##### VM Mode Consistency

A Lite Mode node and a Full Mode node could execute identical logic and diverge.

RESOLVED – SHARED HOST API ENFORCES BYTE-IDENTICAL OUTPUTS AT THE API BOUNDARY, INDEPENDENT OF VM BACKEND.

#### RISK 9.2

##### Cash Mode / TPS Stability

High Cash Mode adoption could significantly reduce effective TPS.

RESOLVED – DALI RAISES THE PRIVACY FEE MULTIPLIER TOWARD 10× ABOVE 15% ADOPTION; 50%+ ADOPTION IS ECONOMICALLY SELF-CORRECTING.

#### RISK 9.3

##### Dandelion++ / Mixing Pool Attacks

Adversarial nodes could attempt deanonymisation during stem-phase propagation or poison mixing pools.

RESOLVED – DALI REPUTATION SCORES GATE MIXING/ROUTING PARTICIPATION; MALICIOUS SUBMISSIONS TRIGGER POEC-STYLE SLASHING.

RISK 9.4

#### RISC-V AOT / Block Production Contention

AOT compilation could compete with TxPoW validation for CPU resources.

RESOLVED — DALI SCHEDULES AOT COMPILATION DURING IDLE PERIODS ONLY; PERSISTENT CACHE ACROSS RESTARTS.

RISK 9.5

#### Stability Mechanism Contention

Multiple stability mechanisms triggering simultaneously could create conflicting actions.

RESOLVED — STRICT PRIORITY HIERARCHY,  $\leq 3$  MECHANISM ACTIVATIONS/HOUR, ADTMS  $\leq 2$  STATE CHANGES/24H.

RISK 9.6

#### QoEx Buyback Market Stability

Large PVT-triggered buybacks could destabilise the QOL/PYM AMM pool.

RESOLVED — BUYBACKS CAPPED AT 10% OF SRP\_PYM/HOUR; THREE-LAYER TWAP PREVENTS FRONT-RUNNING.

RISK 9.7

#### CEA Abuse / Anomaly Detection

SEL nodes could submit CEAs with impossible nonce counts.

RESOLVED — DALI FLAGS SUBMISSIONS EXCEEDING HARDWARE CLASS MAXIMUM THROUGHPUT; PARTIAL PUF BOND CELL SLASHING TO SRP.

RISK 9.8

#### EQCF / ADTMS Signal Manipulation

An adversary could attempt a flash-loan-funded liquidity spike to manipulate  $\hat{L}_t$ .

RESOLVED — 48-BLOCK TRIMMED MEAN, 3-CYCLE EQCF PERSISTENCE GATE, INDEPENDENT 2-CYCLE ADTMS CSI PERSISTENCE GATE — 36-HOUR MINIMUM SUSTAINED COST.

Source: [WP §9.1-9.8] .

05

## Token Accounting — QOL & PYM

### 5.1 QOL — Energy-Backed Flatcoin

QOL is demand-responsive; it has no fixed supply cap. Minting calibration is a seven-stage hardened pipeline composing five independently normalised signals (energy cost, hashrate, velocity, QoEx liquidity depth, volatility) into a confidence-weighted, persistence-gated, hard-bounded scalar [WP §6.1] . Full formula in Section 06.3.

### 5.2 PYM — Fixed-Supply Governance Token

Hard cap: **100,000,000 PYM from genesis**. No additional minting is possible under any protocol state [WP §6.2] . Three foundational UTXO-native Cell types: Governance Cell, Staking Cell, PoEC Bond Cell, each with ML-DSA Level 3 lock scripts by default; extended into the full 16-entry PCTS registry [WP §6.2, §6.7] . Slashed PYM routes to the SRP Buyback Reserve, never destroyed [WP §6.2] .

### 5.3 Fee Routing

| FEE TYPE            | ROUTING                                   |
|---------------------|-------------------------------------------|
| QOL fee (Cash Mode) | 100% → Wormhole burn                      |
| PYM fee (Cash Mode) | 50% → RRP · 50% → SRP Buyback Reserve     |
| PoEC slash          | 100% → SRP Buyback Reserve (never burned) |

Source: [WP App. A] .

5.4 The Pim Cell Type Standard (PCTS)

16 native cell types (PCT-1 through PCT-16) — fully UTXO-native governance, parallel-executable, MMR-provable — spanning payments, identity, escrow, real-world assets, cooperative finance, agent delegation, DAO governance, diaspora remittance, and personal security. Expanded from an original three-cell-type foundation (Governance, Staking, PoEC Bond) [WP §2, §6.7, §12.1] .

06  
Core Algorithm Pseudocode

REPRODUCED FROM WHITEPAPER APPENDIX A

6.1 Sharding — Q-Learning Rebalancing

[WP APP. A]

$Q(s,a) \leftarrow Q(s,a) + \alpha \cdot [R + \gamma \cdot \max_{a'} Q' - Q]$  // R = TPS/Latency - Gini

6.2 PoEM Block Weight

[WP APP. A]

Weight = BLAKE3-XOF(Block Header, Nonce, Timestamp)  
// Minimum weight among candidates wins the slot, deterministically

6.3 EQCF — Seven-Stage Minting Pipeline

[WP §6.1, APP. A]

$\kappa_t = 0.55 \cdot \hat{E}_t + 0.20 \cdot \hat{H}_t + 0.10 \cdot \hat{V}_t + 0.10 \cdot \hat{L}_t + 0.05 \cdot \sigma_t$   
 $EQCF_t = \lambda \cdot \kappa_t + (1-\lambda) \cdot EQCF_{(t-1)}$  //  $\lambda = 0.15$   
 $EQCF_t \in [0.40, 2.50]$  // hard bound, type-script-enforced  
 $\Delta(EQCF_t) \leq 3\%$  per cycle (1.5% in TIGHTENING) // rate cap  
// Persistence: 3 consecutive cycles required for threshold crossing  
  
 $QOL_{minted} = \min(\text{effective\_signal}(EQCF_t, ADTMS.state) \times E_{agg\_t}, \beta \times HR_t) \times F_t$   
effective\_signal: STABLE =  $1.00 \cdot EQCF_t$  | TIGHTENING =  $0.50 \cdot EQCF_t$  | TRANSITION = 0 | CRITICAL = 0  
 $F_t = \text{clip}((TPS_{realized} / 15,100) \times (V_{7d} / V_{hist}), 0.5, 1.5)$  // DemandFactor

6.4 QoEx Liquidity Depth Signal & ADTMS Composite Stress Index

[WP §6.3, APP. A - PIP-016]

$\hat{L}_t = \text{normalise}(\text{trimmed\_mean}_{48\text{block}}(QOL_{pool\_reserve} / QOL_{circulating}), \text{median}_{90d})$   
 $\hat{L}_t \in [0.10, 2.00]$  | fallback  $\rightarrow 1.0$  on anomaly | excludes DaFi vault deposits  
  
 $CSI_t = 0.45 \cdot \text{PriceDeviation}_t + 0.25 \cdot (1 - C_{PoEC,t}) + 0.20 \cdot \text{TPSDeficit}_t + 0.10 \cdot \text{EQCFDrift}_{5c}$   
States: STABLE ( $CSI < 0.25$ ) | TIGHTENING ( $0.25 \leq CSI < 0.50$ ) | TRANSITION ( $0.50 \leq CSI < 0.75$ ) | CRITICAL ( $CSI \geq 0.75$ )  
// Upward transition: CSI above threshold for 2 consecutive cycles  
// Recovery: 6 conditions sustained 4 consecutive cycles, incl.  $C_{PoEC,t} \geq 0.85$   
// Graduated descent invariant: no state-skipping on recovery  
// Transition throttle:  $\leq 2$  state changes per 24h window

6.5 PoEC Slashing & PYM Cell Validation

[WP APP. A]

```
Slash_amt = staked * pct // pct ∈ {0.05, 0.10, 0.15 + 30-day suspension}
SRP_Buyback_Reserve += Slash_amt // never burned

Cell_valid = verify_lock(cell.lock_script, tx.sig) AND verify_type(cell.type_script, tx.inputs, tx.outputs)
```

6.6 PIM-VM Mode Selection

[WP APP. A]

```
mode = if RAM < 4 GB || is_risc_v || power_w < 50 : Lite else : Full
// SEL: if RAM < 512 MB || is_ultra_constrained : SEL (subset of Lite)
```

6.7 Cash Mode Privacy Fee & TPS Impact

[WP APP. A]

```
privacy_fee = base_fee * multiplier // multiplier ∈ [5, 10], DALI-adjusted
QOL_fee → Wormhole_burn(100%)
PYM_fee → RRP(50%) + SRP_Buyback_Reserve(50%)

effective_TPS = base_TPS * (1 - cash_pct * overhead_factor * (1 - dali_gain))
overhead_factor ≈ 0.8 | dali_gain ≈ 0.3
```

6.8 PVT Buyback & QoIEx AMM Invariant

[WP APP. A]

```
Buyback = 0.5 * SRP_PYM * (Vol - 0.018) / 0.1 // capped at 10% SRP_PYM per hour
k = QOL_reserve * PYM_reserve // QoIEx constant-product AMM invariant
```

6.9 DALI Reputation & CEA Reward

[WP APP. A]

```
Rep_i = (Energy_i / ΣE) * Validation_Rate_i * PoEC_Honesty_i

CEA_Reward = base_rrp_rate * hardware_class_factor * honesty_score
hardware_class_factor: SEL-0 = 0.05 | SEL-1 = 0.15 | SEL-2 = 0.35 | Full PoEC = 1.0
```

6.10 PUF Key Derivation & Calibrated Energy

[WP APP. A]

```
stable_seed = fuzzy_extractor(raw_puf_response)
signing_key = ml_dsa::keygen_from_seed(shake256(stable_seed, "pim-sel-signing-key-v1"))

calibrated_energy = txpow_nonce_count * hardware_class.energy_per_op
```

07  
Parameter Tables

7.1 Performance Benchmarks

| Metric                       | Target / Specification |
|------------------------------|------------------------|
| Throughput (theoretical max) | 100,000 TPS            |

|                                           |                                                     |
|-------------------------------------------|-----------------------------------------------------|
| Throughput (sustained baseline)           | 15,100 TPS                                          |
| Throughput (heterogeneous fleet ceiling)  | 60,000-80,000+ TPS                                  |
| Confirmation latency                      | 0.3-0.5s intra-shard · ~8-10ms owned-cell fast-path |
| Cross-shard finality                      | ~8s (DFP)                                           |
| Fork probability (10,000 nodes, sim)      | 0.0007%                                             |
| Energy per transaction                    | 0.00034 kWh                                         |
| QOL volatility target                     | 1-2%                                                |
| ML-DSA Level 3 signature                  | 0.1ms · pk 1,952B · sig 3,309B                      |
| SLH-DSA signature                         | 0.2ms (backup/Wormhole)                             |
| DALI volatility/fraud prediction accuracy | ≥97% (target)                                       |

Source: [WP §4.2, §11]

7.2 Node Hardware Requirements

| TIER                      | RAM / STORAGE / POWER       | VM MODE                    | PRIMARY ROLE                                      |
|---------------------------|-----------------------------|----------------------------|---------------------------------------------------|
| SEL (Tier 0)              | 256KB-1.5GB / any / 0.1-8W  | SEL Mode (wasm3 + PUF CEA) | TxPoW · own-coin MMR · CEA · Lite DALI            |
| Edge Coordinator (Tier 1) | 512MB-1.5GB / 8GB / 5-15W   | Lite + TrustZone/PUF CEA   | SEL cluster aggregation · light mixing            |
| Mobile/IoT (Tier 2)       | 1.5-3GB / ≥8GB / 8-40W      | Lite (Full if ≥4GB)        | Lite PoEC-NFT · TxPoW · KISS Cell ops             |
| PC (Tier 3)               | 4-8GB / 256GB SSD / 50-100W | Full (wasmtime)            | Full PoEC-NFT · DALI aggregation · cross-shard    |
| GPU (Tier 4)              | ≥16GB / ≥512GB SSD / ~400W  | Full (wasmtime + lazy-JIT) | Full PoEC-NFT · block production · ZKP generation |

Source: [WP App. B]

7.3 EQCF / ADTMS / PoEC Cycle Parameters

| PARAMETER                    | VALUE                                                         |
|------------------------------|---------------------------------------------------------------|
| EQCF settlement cycle        | 12h, confidence-weighted EMA                                  |
| EQCF max drift               | 3%/cycle (1.5% TIGHTENING)                                    |
| EQCF hard bounds             | [0.40, 2.50]                                                  |
| ADTMS recovery requirement   | 6 conditions, 4 consecutive cycles, 85% PoEC compliance floor |
| PoEC cycle (Tier 2-4)        | 12h – Full PoEC-NFT                                           |
| CEA cycle (Tier 0-1)         | 12h – PUF-CEA                                                 |
| QolEx liquidity signal (L_t) | [0.10, 2.00], STARK-committed every 4 min                     |

Source: [WP §4.2, §11, App. A]

08  
Spec Gaps

DETAILS REQUIRED FOR FULL IMPLEMENTATION NOT PRESENT IN THE WHITEPAPER

|                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Spec gap:</b> QOL genesis/initial circulating supply figure. QOL is explicitly demand-responsive with no fixed cap [WP §6.1] , but no starting supply value is given.                                                                     |
| <b>Spec gap:</b> PYM token generation event structure — allocation across team, treasury, ecosystem, and public participants; any vesting or lockup schedule. Only the 100,000,000 hard cap is specified [WP §6.2] .                         |
| <b>Spec gap:</b> Mainnet launch date. Only a testnet milestone is referenced: "Open Engineering Item — Q3 2026 Testnet" [WP §8, near §8 close] .                                                                                             |
| <b>Spec gap:</b> Formal third-party security audit report or auditor identity. The Whitepaper describes internal STARK-commitment and ML-DSA-signing audit trails for monetary events [WP §7] but does not reference an external code audit. |
| <b>Spec gap:</b> PimBridge cross-chain specification. Referenced only as "(planned)" for EU market interoperability, with no architecture detail [WP §10.2] .                                                                                |
| <b>Spec gap:</b> Full field-level schema for all 16 PCTS Cell types beyond the summary in §6.7 and the 3-type foundational description in §2 — detailed struct definitions are out of this Whitepaper's scope.                               |
| <b>Spec gap:</b> Exact PYM governance quorum thresholds and proposal lifecycle timing (voting period length, proposal bonding requirements) beyond the general UTXO-native governance description [WP §6.2, §6.7] .                          |
| <b>Spec gap:</b> DALI model architecture hyperparameters (LSTM layer count, training data retention policy) beyond the stated ≥97% accuracy targets and <10KB model-delta distribution mechanism [WP §8] .                                   |

## Source Mapping & Notice

This Bluepaper's section-to-Whitepaper mapping: §02 ← WP §4.1, §5. §03 ← WP §6.3. §04 ← WP §4.1, §9, §11, §12.1. §05 ← WP §6.1, §6.2, §6.7. §06 ← WP Appendix A. §07 ← WP §4.2, §11, Appendix A, Appendix B. §08 is derived by comparison against the full Whitepaper text; no external source.

**Notice.** This document is a derived technical specification prepared by Pim Global Limited and Qollabs Ltd from the Pim Protocol Technical Whitepaper. Performance figures represent design targets and simulation results; mainnet figures may vary. Pim Protocol has not yet reached mainnet launch as of the date of this document. This document does not constitute an offer to sell or a solicitation of an offer to buy any security or financial instrument in any jurisdiction. QOL and PYM are utility and governance instruments of the protocol described herein, not investment contracts.

### Pim Protocol

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